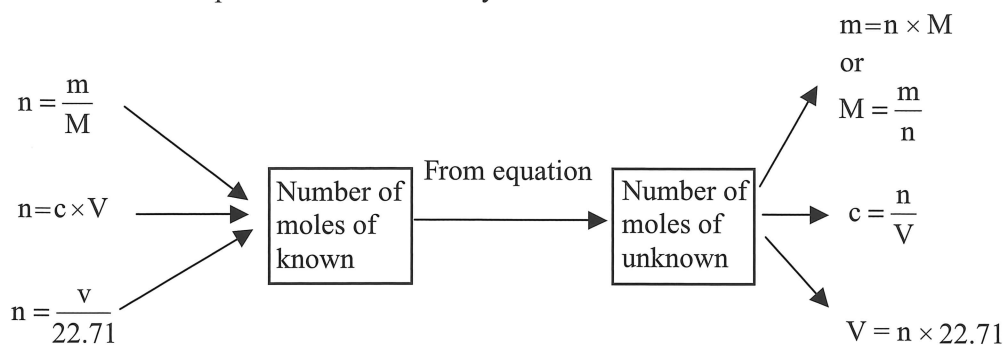


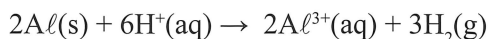
Set 33: Reacting masses and gaseous and solution volumes

Using the stoichiometric relationship between the numbers of moles of reactants and products in a chemical reaction, a range of calculations can be carried out involving the masses, gaseous volumes and solution volumes of reactants and products. These calculations can be represented schematically as follows:



Set 33: Exercises

- 25.0 mL of $0.0227 \text{ mol L}^{-1}$ silver nitrate solution is added to excess sodium chloride solution.
 - Write an equation for the reaction.
 - Calculate the mass of the precipitate formed.
- 47.3 L of butane (C_4H_{10}) measured at S.T.P. is burned in excess oxygen.
 - Write an equation for the reaction.
 - What volume of carbon dioxide (measured at S.T.P.) is produced?
 - What mass of water is produced?
- 1.34 g of solid sodium carbonate is added to excess 0.125 mol L^{-1} hydrochloric acid solution.
 - Write an equation for the reaction.
 - What volume of CO_2 (measured at S.T.P.) is produced?
 - What volume of hydrochloric acid would be required to react completely with the sodium carbonate?
- Magnesium reacts with dilute hydrochloric acid according to the equation
$$\text{Mg(s)} + 2\text{H}^+(\text{aq}) \rightarrow \text{Mg}^{2+}(\text{aq}) + \text{H}_2(\text{g})$$
If 0.720 g of magnesium is treated with 0.950 mol L^{-1} hydrochloric acid, calculate:
 - the volume of hydrochloric acid needed to react with all the magnesium
 - the volume of hydrogen gas formed at S.T.P.
 - the mass of magnesium chloride which would be produced in solution
- Hydrogen gas is produced by adding sulfuric acid to aluminium according to the equation



If 19.6 L of hydrogen is produced at S.T.P., calculate:

- the volume of 6.00 mol L^{-1} sulfuric acid required
- the mass of aluminium consumed

Notes

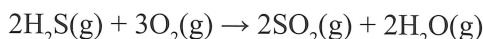
Set 33: Reacting masses and gaseous and solution volumes

Notes

6. (a) What volume of 0.260 mol L^{-1} sodium iodide is needed to precipitate all the lead ions in 25.0 mL of 0.212 mol L^{-1} lead(II) nitrate?
(b) Calculate the mass of lead(II) iodide precipitated.
7. Carbon dioxide can be produced by reacting calcium carbonate with hydrochloric acid. If 10.1 g of calcium carbonate is treated with 1.07 mol L^{-1} hydrochloric acid, calculate:
(a) the volume of HCl needed to completely react with the CaCO_3
(b) the volume of CO_2 produced at S.T.P.
(c) the concentration of calcium chloride in the final solution, assuming no change in the volume of solution
8. A calcite crystal (CaCO_3) with a mass of 5.60 g is placed in a beaker with 20.0 mL of dilute hydrochloric acid. The solution is warmed gently until the reaction is complete. The remaining calcite is removed and dried and found to have a mass of 5.09 g . Calculate the concentration of the hydrochloric acid.
9. An excess of hydrochloric acid was added to a sample of silver carbonate. After the reaction the silver chloride residue was washed and dried and found to have a mass of 5.74 g :
- $$\text{Ag}_2\text{CO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow 2\text{AgCl}(\text{s}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$$
- Calculate
(a) the mass of the silver carbonate sample
(b) the number of moles of the acid consumed
(c) the volume of carbon dioxide gas, measured at S.T.P., given off during the reaction
10. When white phosphorus is heated in air it burns to form tetraphosphorus decaoxide (P_4O_{10}):
- $$\text{P}_4(\text{s}) + 5\text{O}_2(\text{g}) \rightarrow \text{P}_4\text{O}_{10}(\text{s})$$
- If 6.20 g of phosphorus is treated in this way:
(a) what mass of P_4O_{10} is formed?
(b) what volume of air (measured at S.T.P.) is required?
Assume that air is 20.0% oxygen by volume.
11. A 15.0 g piece of zinc is added to a beaker containing dilute hydrochloric acid and effervescence occurs:
- $$\text{Zn}(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{ZnCl}_2(\text{aq}) + \text{H}_2(\text{g})$$
- or
$$\text{Zn}(\text{s}) + 2\text{H}^+(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{H}_2(\text{g})$$

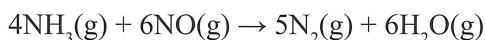
After all reaction has ceased, the piece of zinc is washed, dried and found to have a mass of 2.00 g . Deduce from these results
(a) the number of moles of hydrochloric acid in the beaker
(b) the mass of zinc chloride which can be recovered from the solution in the beaker
(c) the volume of hydrogen gas produced by the reaction at S.T.P.

12. It has been estimated that 1.00×10^8 tonne of H_2S is released annually into the atmosphere from biogenic sources, especially from sulfate respiration by bacteria. This H_2S is oxidised to SO_2 in a day or so according to the overall process:



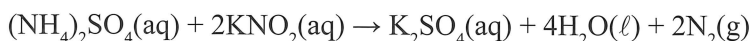
If 8.00 g of oxygen is consumed by a sample of hydrogen sulfide;

- (a) how many mole of hydrogen sulfide molecules would react?
(b) what volume at S.T.P. of hydrogen sulfide would react?
13. A technique, known as the Nanaimo system for reducing the NO content of car exhausts involves the injection of a stream of NH_3 into the exhaust vapour. This converts the NO to harmless N_2 and H_2O via the reaction



Cars typically emit about 3.00 g of NO per km travelled. How many litres of NH_3 , calculated at S.T.P., would be needed to clean up the NO emitted by a car in one year, assuming that it is driven a distance of 1.60×10^4 km?

14. Liquid hydrogen is being used to power an experimental jet engine. In a particular test, in which all of the hydrogen fuel was oxidised, the fuel tank was found to have decreased in mass by 2.00 kg. What volume of oxygen, measured at S.T.P. would be needed in this test run?
15. 1.60 g of hydrogen gas is added to a container with 10.0 L of oxygen gas at S.T.P. The mixture is sparked and a reaction occurs. Find:
- (a) the mass of steam formed;
(b) the volume of unused oxygen at S.T.P.
(c) the volume of steam produced at S.T.P.
16. What volume of oxygen is required for the complete combustion of
- (a) 1.00 L of hydrogen at S.T.P.?
(b) 2.00×10^2 mL of methane (CH_4) at S.T.P.?
17. When 25.0 mL of a solution of dilute sulfuric acid is mixed with excess barium chloride solution, 0.483 g of precipitate is formed. What volume of $0.0134 \text{ mol L}^{-1}$ sodium hydroxide solution is required to neutralise 125 mL of the original acid?
18. A 3.00 g sample of limestone is analysed for purity by reacting it with hydrochloric acid. It is found that 20.0 mL of 2.50 mol L^{-1} HCl solution is required for complete reaction. Calculate the percentage of calcium carbonate in the limestone sample.
19. A 0.482 g sample of an alloy of silver and copper was dissolved in nitric acid. The resulting solution required 41.5 mL of $0.0993 \text{ mol L}^{-1}$ sodium chloride solution for complete precipitation of the silver ions. Calculate the percentage of silver in the alloy?
20. A solution containing 30.0 g of ammonium sulfate and 34.0 g of potassium nitrite is heated and the following reaction occurs:



- (a) Which reactant is used up?
(b) What mass of the other reactant remains?
(c) What volume of nitrogen gas, measured at S.T.P., is evolved?